

1. Intro. (FSA, automatic groups, Stallings $\mathbb{F}_n$) (not in depth, link if more needed)
2. Stallings automatic gps KMW'17
3. Stallings automatic gps : subnormality.
4. Application to surface groups.

1. Intro

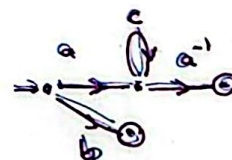
1.1. Notation:

- S finite alphabet closed under inverses $S = \{a, b, c, a^{-1}, b^{-1}, c^{-1}\}$
(f.e. gener. set for group)
- S^* denotes set of words over S $S^* = \{ \epsilon, a, ab, ac^{-1}, cb, \dots \}$
- $\mathbb{F}(S)$ is free gp over S .
- A language L is any subset of S^*
- All groups are f.p. $G = \langle S | R \rangle$ with projection $\pi: S^* \rightarrow G$

1.2. FSA

Def: A FSA A over S is a fte, dir. graph with:

- S -labelled edges
- A distinguished start vertex $\rightarrow \bullet$
- A set of accepting vertices $\odot$



The language accepted by A , $L(A) \subseteq S^*$, is the set of labels of paths from $\rightarrow \bullet$ to $\odot$ in A .

$$L(A) = \{ aa^{-1}, aca^{-1}, acca^{-1}, b, \dots \}$$

A language $L \subseteq S^*$ is regular if $L = L(A)$ for some A FSA.

We use FSA to

- 1) define groups (automatic groups)
- 2) define algorithms in groups (Stallings automata)

1.2. Automatic groups

Def : An automatic group is a f.g. group $G = \langle S \rangle$ equipped w/:

1. A regular language $L \subseteq S^*$ s.t. $\pi(L)$ surjects onto G . Rational structure
2. A set of FSA $\{M_s\}_{s \in S \cup \{id\}}$: multipliers

Eg: Hyperbolic groups and RAAGs are automatic. $BS(n, m)$ are not.

1.3. Stallings automata

Def: Given $X \subseteq G = \langle S \rangle$, the membership problem for X in G

is decidable if, given $w \in S^*$, we can say whether $\pi(w) \in X$
(= understand $\pi^{-1}(x)$)

The subgroup / submonoid membership problem in G is decidable

if the membership problem is decidable for all f.g. subgp / subm. in G .

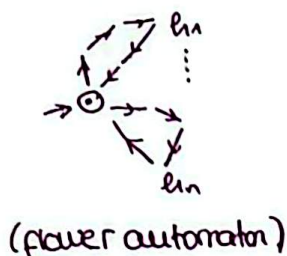
Def: Stallings '83 defined iterative procedure, given $H \leq_{fg} F(S)$,
to build FSA A s.t. $L(A) = \{\text{freely reduced words in } H\}$.

The automaton is Stallings automaton, procedure is Stallings folding

& it solves subgroup membership problem of $F(S)$.

Construction: Given $H = \langle e_1, \dots, e_n \rangle \leq F_S$

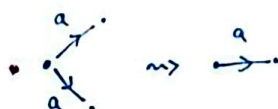
① Build FSA A_0



② "Fold" A_0

$A_0 \rightarrow A_1 \rightarrow A_2 \dots$

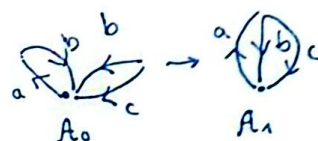
perform edge identification



③ (?) ④ Condition to stop

stop at A_n when
no more foldings
possible.

$H = \langle ab, b^{-1}c \rangle$



- A_n solves ~~the~~ membership problem for H .

- and more!

- very simple, yet very powerful.

- Afterwards many generalisations: same structure,
other groups.

Ben-zvi, Kropholler, Lyman'11: Stallings for π_1 (npc cc)

← Dani, Levicontz '21: Stallings for ~~some~~ RACG

Khorlamponich, Miasnikov, Weir'17: for automatic groups

Always
pay
price ...

This is the topic for today!

For rest of talk: $G = \langle S | R \rangle$ automatic,
f.p.

L rational structure, 1st & multipliers.

Why automatic groups? Here, membership problem for $\chi \equiv$

understanding $\pi^{-1}(\chi) \equiv$

understanding $\pi^{-1}(\chi) \cap L$

" L -representatives" of χ .

Next: KMW'17 + our generalisation to submonoids + an application to surface gps.

2. Khorlamovich, Miasnikov, Weil '17

Gave Stallings-like procedure for subgroup membership in automatic groups. $G = \langle S | R \rangle$

① Take $H = \langle h_1, \dots, h_n \rangle \leq G$

Build FSA $A_0 =$ 

(flower)

$\pi(L(A_0)) = H$,

but it does not solve membership prob!!

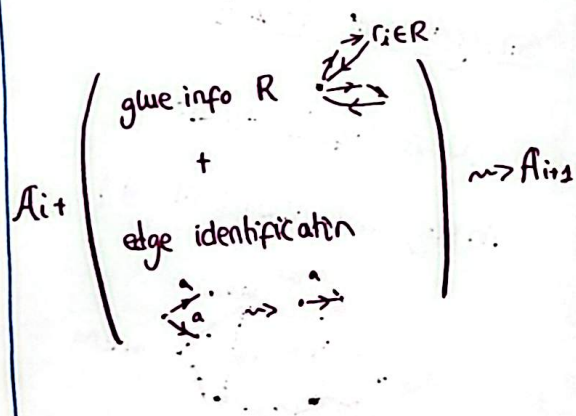
② "Fold" A_0 . Describe

$A_0 \rightarrow A_1 \rightarrow \dots$ s.t.

- $L(A_i) \subseteq L(A_{i+1})$

- $\pi(L(A_i)) = H$

- Accept more L-reps of H
 $L \cap \pi^{-1}(H)$



③ Have geometric condition:

H satisfies condition $\Rightarrow$ ^{Gersten-Sat '92}

$\exists i_0: L(A_{i_0}) \geq L \cap \pi^{-1}(H)$

Condition is "L-quasiconvexity".

H is L-qc if $\exists K$ s.t.:

In $\text{Cay}(G, S)$, $\forall w \in L \cap \pi^{-1}(H)$



$\exists h_1, \dots, h_{l(w)} \in H: d(w_i, h_i) \leq K$

④ Have test to detect i_0 & stop algorithm.

use multipliers $\{R_i\}$.

Solves subgroup membership problem for L-qc subgroups of automatic groups.

3. Our approach

Generalise KMW'14 to work with submonoids of automatic groups.

Alter ②, ③ strengthen condition in ③.

① Take $\mu = \langle t_1, \dots, t_n \rangle^+ \leq G$
subm

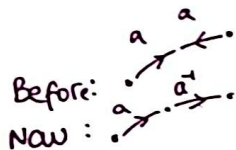
Build FSA A_0



(flower)

$$\pi(L(A_0)) =$$

$$\pi(\{t_1, \dots, t_n\}^*) = M$$



② "Fold" A_0 . Describe

$$A_0 \rightarrow A_1 \rightarrow \dots \text{ s.t. :}$$

$$- L(A_0) \subseteq L(A_1)$$

$$- \pi(L(A_i)) = M$$

$$- L(A_i) \text{ more } L\text{-reps of } \mu$$

$$A_i + \left(\begin{array}{l} \text{glue info } R \text{ in } A_i \\ + \\ \text{modify } A_i \text{ to accept} \\ \text{words up to free reduction} \\ + \\ \text{determinisation} \\ \text{(instead of edge identis)} \end{array} \right) \rightarrow A_{i+1}$$

③ Have geom condition:

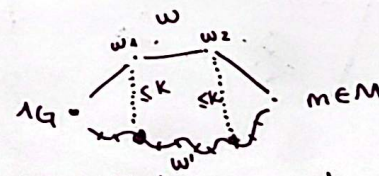
H satisfies condition $\Rightarrow$
 $\neq!$

$$\exists i_0 : L(A_{i_0}) \geq L \cap \pi^{-1}(M)$$

Strengthened L-gc:

μ is L-gc if $\exists K$ s.t:

In $\text{cay}(G, S)$, $\forall w \in L \cap \pi^{-1}(M)$



$$\exists w' \in L(A_0) : w =_G w' \text{ and}$$

w' asynchronously K -fellow travels with w $m_i \in m_i M$

④ Have test to detect io
& stop algorithm.

Same as KMW'17.

this is for free in L-gc subgroups
can be related more similar to KMW'17

$$\exists m_1, \dots, m_i, m_i \in M$$

$$d(m_i, w_i) \leq K$$

Solves

submonoid membership problem for L-gc submonoids of automatic groups

Remarks

- we originally defined algorithm for rational subsets
(for stronger L-qc.), but we do not have test here.
 $\downarrow$
controlling level asynchronicity
 - If you know K constant of L-qc, you do not need test
(you can estimate λ_0)
 - Automaticity only used in test, folding only requires
$$\begin{matrix} L & \xrightarrow{\quad} & G \\ \text{N reg} & & \langle P \leq S \rangle \\ S^* & & \end{matrix}$$
- Feasible to have groups where test can be built in other ways.
- Strengthened L-qc was defined to fit the algorithm,
what submonoids are L-qc? we need examples.

4. Application to surface groups

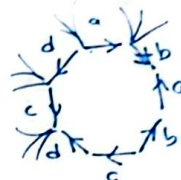
Goal: find L-qc submonoids of surface groups.

Def: The surface group of genus g

$$\Sigma_g = \pi_1 \left(\underbrace{\text{closed, compact, orientable manifold}}_{g \geq 2} \right) = \langle \underbrace{a_1, b_1 \dots a_g, b_g}_{S \text{ from row on}} \mid \underbrace{[a_1, b_1] \dots [a_g, b_g]}_{\pi} = 1 \rangle$$

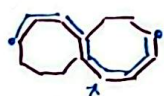
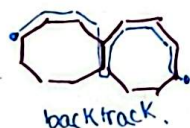
Why look at them?

- Σ_g is geodesically automatic (L-geodesics in $\text{Cay}(G, S)$)
- $\text{Cay}(G, S)$ gives a tiling of H^2 by $4g$ -gons
- $\langle S | r \rangle$ is a special presentation (Dehn 1912, English 1987) (star Dehn rewriting system associated, strong small-cancellation properties)

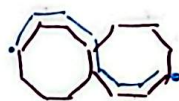


Def: $\Sigma_g = \langle S | r \rangle$. we say $w \in S^*$ is Dehn-reduced if, as a path in $\text{Cay}(\Sigma_g, S)$, w has no backtracking ($\cdot \rightarrow \cdot \rightarrow \cdot$) and does not go around more than half of a $4g$ -gon ()

Eg:



5/8 of octagon



Dehn-reduced

Remark

Dehn-reduced $\Rightarrow$ geodesic.

Almost!

Thm: McCammond-Wise '02. Fans and ladders... applied to Σ_g :

$\Sigma_g = \langle S | r \rangle$. Let $w \in S^*$ geodesic in $\text{Cay}(\Sigma_g, S)$

Let $w' \in S^*$ Dehn-reduced with $w =_G w'$

Then: a Van Kampen diagram for $w(w')^{-1}$ is a "ladder", in particular:

w' asynch. $(g+1)$ -fellow travels with w .

Cor: $T = \{t_1 \dots t_n\} \leq S^*$. If T^* is Dehn reduced, the monoid $\langle T^+ \rangle \leq \Sigma_g$ is L-qc and has decidable membership problem.

Construction: Given $\Sigma_g = \langle S \mid r \rangle$, take $T \subseteq S^*$ s.t.:

① $\forall t \in T$, t is Dehn-reduced and $|t| \geq g$

Avoids $t_i t_j$ goes around $> \frac{1}{2}$ 4g-gon.

② $\forall t \in T$, prefix (t, g) is not subword of r, r^{-1} or a cyclic permutation
" suffix (t, g) " " " " " " " " " " " "

Avoids backtracking $t_i t_j$

③ Let $T[0] := \{ \text{set of first words letters of } t \in T \}$

$T[-1] := \{ \dots \text{last} \dots \}$

s.t. $T[0] \Rightarrow s^{-1} \notin T[-1]$

Claim: T^* is Dehn-reduced

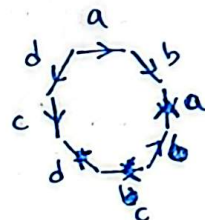
($\Rightarrow$ submonoid $\langle T^* \rangle^+$ is L-quasi-convex)

Eg. $g=2$

$w_1 = \underline{ac} v_1 \underline{c^{-1}d}$, $v_1 \in S^*$ Dehn-red.

$w_2 = \underline{ad^{-1}}$

$w_3 = \underline{bc} v_3 \underline{bd}$ $v_3 \in S^*$ Dehn-reduced



$\{w_1, w_2, w_3\}^*$ is Dehn-reduced

$w_4 = d^4 c d a$

$\{w_1, w_2, w_3, w_4\}^*$ is not Dehn reduced!

$w_2 w_4 = a d^5 d c d a$
($=_{\Sigma_2} a c d a$)

Remark

we could be more relaxed... $\{w_1, w_2, w_3, w_4\}$ is not Dehn reduced but is very close from being! We don't need $g+1$ -fellow traveller.

we could ask for T^* to be at most M "simultaneously-Dehn-reduction" away from being Dehn reduced,

and $\langle T \rangle^+$ would still be L-qc.